SYSTEM AND METHOD FOR IMAGING AN INTERIOR SPACE USING A PAIR OF COORDINATED MOBILE TRANSCEIVER UNITS

Reinaldo Javier Aguiar Marcano

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Title: SYSTEM AND METHOD FOR IMAGING AN INTERIOR SPACE USING A PAIR OF COORDINATED MOBILE TRANSCEIVER UNITS

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A system and method for remotely imaging the interior of a target space, such as a building or vehicle, using a pair of coordinated mobile units. The system comprises a first mobile unit equipped with a signal emitter (transmitter) and a second mobile unit equipped with a synchronized signal sensor (receiver). The two units are positioned on opposing sides of a target space. The transmitter emits a non-ionizing radiation signal (e.g., millimeter-wave) which penetrates the walls or fuselage of the target space. The receiver captures the attenuated signal that has passed through the interior of the space. By analyzing the differential attenuation of the signal, a two-dimensional or three-dimensional image of the contents within the target space, including metallic objects or human occupants, can be computationally reconstructed. The system is designed for rapid, non-invasive inspection of structures from a standoff distance.

Background of the Invention

The non-invasive inspection of sealed structures, such as buildings or vehicles, is a significant challenge in the fields of security, surveillance, and law enforcement. Technologies for "seeing through walls" have been the subject of extensive research. Current methods include backscatter X-ray systems, which are large, costly, and involve ionizing radiation, making them unsuitable for many field applications. Other technologies, such as ground-penetrating radar, are optimized for subsurface imaging and are less effective at resolving details within an above-ground structure from a distance.

There exists a need for a mobile, non-invasive imaging system that can be rapidly deployed to determine the contents or occupancy of a target space from a safe, standoff distance. Such a system should ideally use non-ionizing radiation, be deployable on conventional vehicles, and not require physical access to the target structure. The present invention addresses this need by adapting the principles of transmission imaging to a mobile, two-vehicle system.

Summary of the Invention

The present invention provides a system for imaging the interior of a target structure using two coordinated mobile units, typically vehicles. A first vehicle, the "Transmitter Unit," is equipped with a directional antenna that emits a penetrating, non-ionizing signal, such as millimeter-wave (MMW) radiation. A second vehicle, the "Receiver Unit," is equipped with a synchronized, directional sensor array designed to detect the same signal frequency.

In operation, the two units are positioned on opposite sides of a target structure, such as a house or a vehicle. The Transmitter Unit directs its beam towards the target. The signal penetrates the structure's exterior walls, is attenuated by objects and occupants inside, and exits through the opposite wall. The Receiver Unit, precisely synchronized in time and location with the transmitter, captures the resulting attenuated signal.

A central processing system, which may be located in one of the vehicles or at a remote command center, compares the transmitted signal strength with the received signal pattern. By computationally analyzing the differential attenuation across the signal path, a 2D projection image of the interior space is reconstructed. By moving the vehicles in a synchronized path or by using multiple transmitter/receiver pairs, a 3D tomographic reconstruction can be generated. The system allows for the rapid, covert, and non-invasive determination of a structure's contents or occupancy from a standoff distance.

Brief Description of the Drawings

FIG. 1 is a diagram illustrating the overall system in operation, showing the transmitter vehicle and receiver vehicle positioned on opposite sides of a target house.

FIG. 2 is another view of the system in operation, showing the two vehicles positioned on either side of a target vehicle.

FIG. 3 is a block diagram illustrating the functional components of the transmitter unit, the receiver unit, and the processing system.

Detailed Description of the Invention

Referring to **FIG. 1**, the invention comprises a system of at least two coordinated mobile units for imaging the interior of a target structure (101), such as a building. A first mobile unit, the Transmitter Vehicle (102), is positioned on one side of the target. A second mobile unit, the Receiver Vehicle (103), is positioned on the opposite side.

The Transmitter Vehicle (102) is equipped with a directional signal emitter which generates a penetrating, non-ionizing signal (104), such as a millimeter-wave (MMW) beam. The signal is directed at the target structure (101). The signal penetrates the exterior wall, passes through the interior space, and is attenuated by objects (105) or occupants within the structure before exiting through the opposing wall.

The Receiver Vehicle (103), which is synchronized in time and location with the Transmitter Vehicle, captures the attenuated signal (106) with a specialized sensor array.

As shown in **FIG. 2**, the same principle applies to smaller targets, such as a Target Vehicle (201). The Transmitter Vehicle (102) and Receiver Vehicle (103) position themselves on either side of the Target Vehicle and perform the same scanning operation.

The functional components of the system are detailed in the block diagram of **FIG. 3**. The Transmitter Unit (housed in vehicle 102) comprises a Signal Generator (301), a Directional Antenna (302), a precise Timing & Synchronization module (303) (e.g., GPS-locked clock), and a wireless communication link (304). The Receiver Unit (housed in vehicle 103) comprises a corresponding Directional Sensor Array (305), its own Timing & Synchronization module (306), and a wireless communication link (307).

Both units communicate with a Processing System (308), which may be housed in either vehicle or at a remote location. In operation, the Processing System (308) commands the Transmitter Unit to begin emitting. The signal (104) is generated and transmitted. The

Receiver Unit, being synchronized, captures the resulting attenuated signal (106). The captured signal data is sent to the Processing System (308).

The Processing System (308) performs a computational analysis, comparing the known characteristics of the original signal with the attenuated signal received. By mapping the variations in signal strength across the sensor array, the system's Image Reconstruction Logic (309) generates a 2D or 3D image (310) of the contents within the target structure. To create a 3D image, the vehicles (102, 103) may move in a synchronized path around the target, capturing multiple projections which are then computationally combined in a tomographic reconstruction algorithm.

Claims

What is claimed is:

1. A system for imaging an interior of a target space, comprising:
    - a first mobile unit equipped with a directional signal transmitter configured to emit a signal capable of penetrating the boundaries of the target space;
    - a second mobile unit equipped with a directional signal receiver configured to detect said signal after it has passed through the target space;
    - a synchronization module configured to coordinate the timing and positioning of the first and second mobile units; and
    - a processing system configured to receive data from the signal receiver and to computationally reconstruct an image of objects within the target space by analyzing the attenuation of the signal.
2. The system of claim 1, wherein the first and second mobile units are vehicles.
3. The system of claim 1, wherein the signal is a non-ionizing radiation signal.
4. The system of claim 3, wherein the signal is a millimeter-wave signal.
5. The system of claim 1, wherein the processing system is further configured to generate a three-dimensional tomographic image by combining data from multiple projections captured as the mobile units move in a synchronized path relative to the target space.
6. A method for imaging an interior of a target space, comprising the steps of:

- Positioning a first mobile transmitter unit and a second mobile receiver unit on substantially opposite sides of the target space;
- Transmitting, from the first unit, a penetrating signal through the target space towards the second unit;
- Receiving, at the second unit, the signal after it has been attenuated by the target space and its contents;
- Communicating the received signal data to a processing system; and
- Computationally generating an image of the contents of the target space based on a differential analysis of the signal's attenuation.

7. The method of claim 6, wherein the penetrating signal is a millimeter-wave signal.

8. The method of claim 6, further comprising the step of moving the first and second units in a synchronized path relative to the target space to capture a plurality of projections for tomographic reconstruction.

Drawings

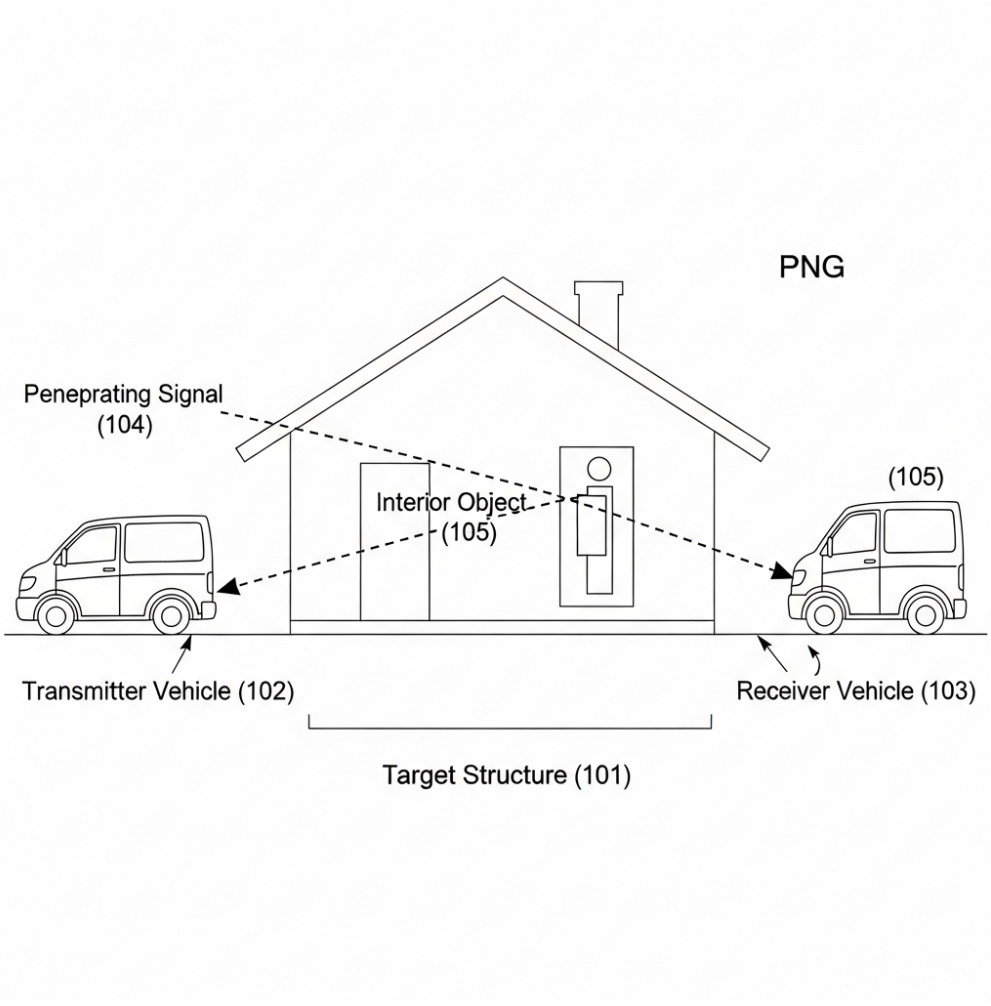


Figure 1: System in operation with a target house.

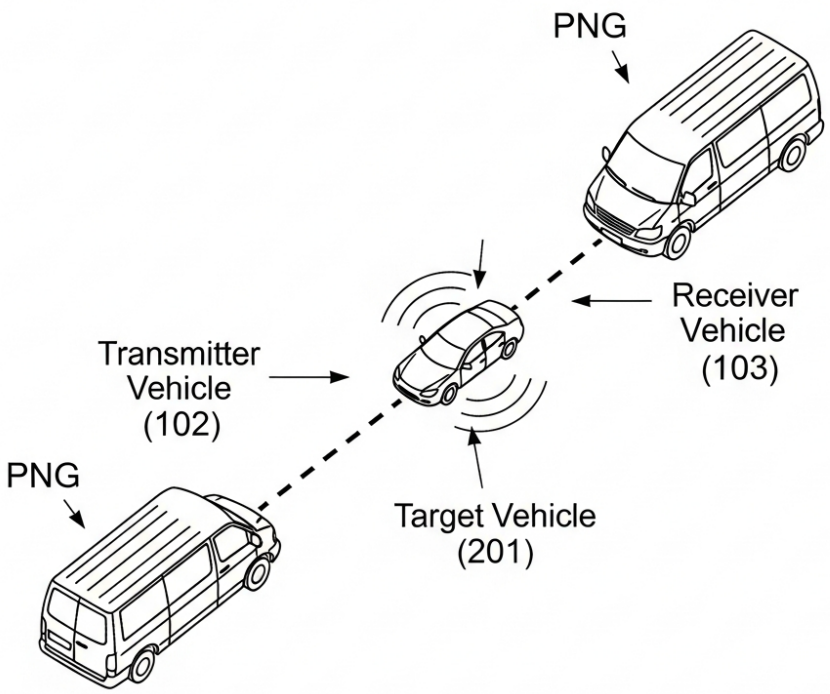


Figure 2: System in operation with a target vehicle.

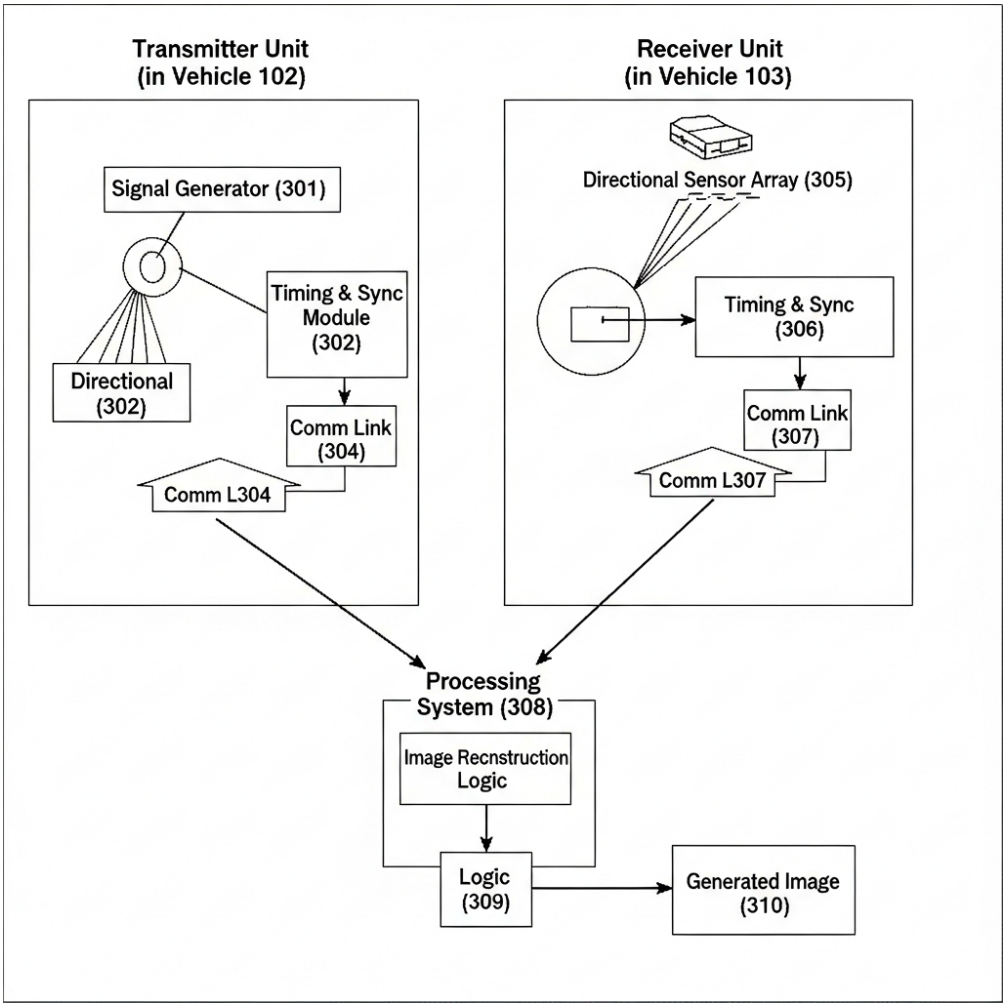


Figure 3: Block diagram of system components.